# LCEC-IO — EtherCAT slave I/O board

Root sheet, milestone M3: Power (2) and MCU (3). Sheets 4 to 8 (EtherCAT, digital inputs, MPG, relay outputs, analog outputs) are added in M4 and M5; the full root sheet follows in M6.

Power

File: 02_power.kicad_sch

MCU

MPG_A
MPG_B
ESC_IRQ
ESC_SYNC0
ESC_SYNC1
ESC_MISO
DI[1..8]
SEL[1..12]
RELAY_OE
ERR_LED
ESC_CS
ESC_SCK
ESC_MOSI
ESC_RST
DAC_CS
DAC_SCK
DAC_MOSI
AO_CLAMP
DO[1..8]
CLK_ESC

File: 03_mcu.kicad_sch

No-connect flags on the MCU sheet pins are placeholders: they are replaced by wires to sheets 4 to 8 in M4 and M5.

Root sheet: milestone M3 holds the Power and MCU sheets; sheets 4–8 follow in M4 and M5

**Yantra Innovation — LCEC-IO**

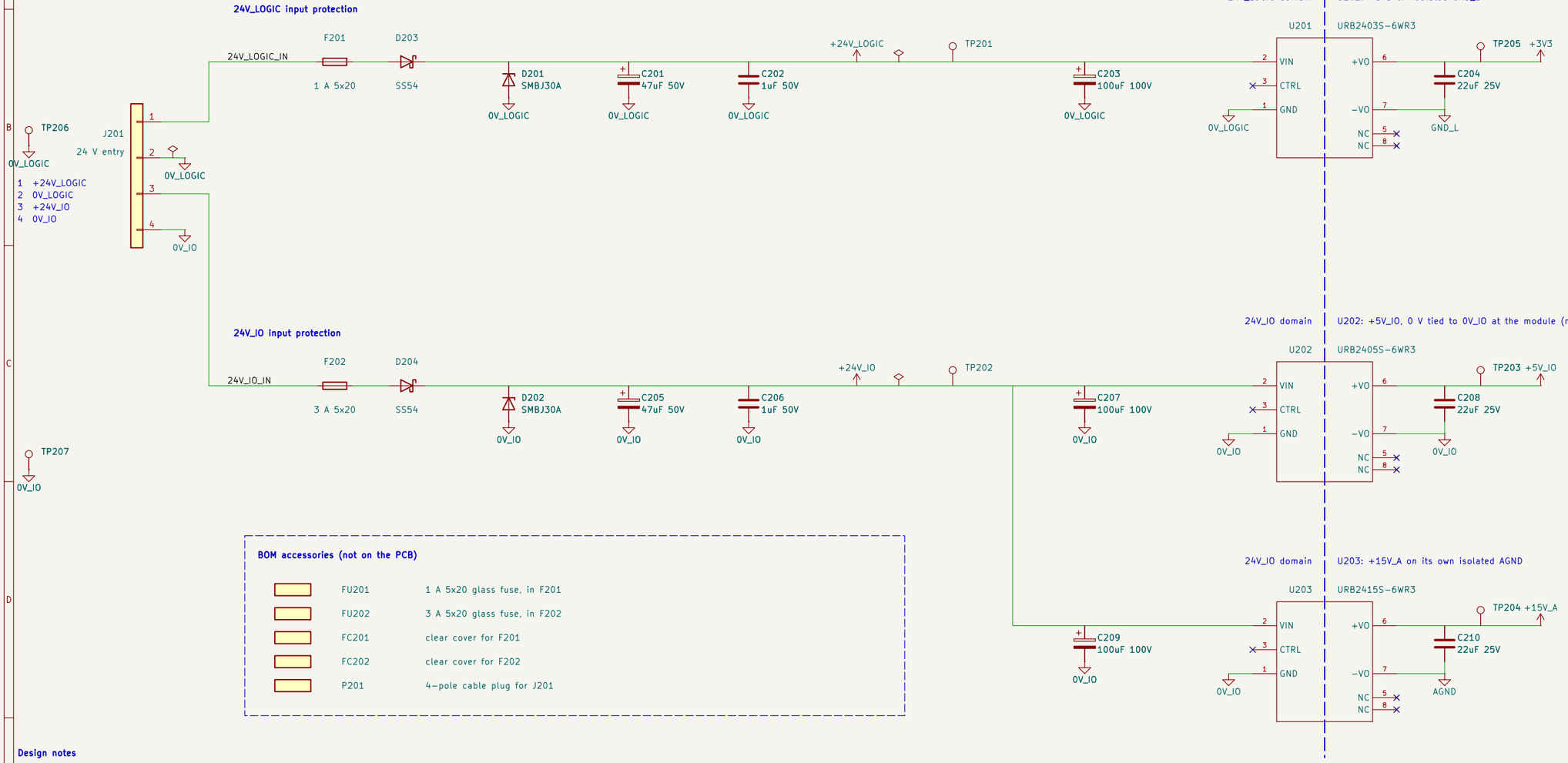
Sheet: /
File: lcec-io.kicad_sch

***Title: LCEC-IO — EtherCAT slave I/O board***

| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 1/3 |

# 2 POWER – 24 V entries, protection, three isolated DC–DC modules

Input 24 VDC +/–20 % = 19.2 … 28.8 V. Per input: fuse > SS54 reverse–polarity diode > SMBJ30A TVS > 47 uF + 1 uF, then a 100 uF/100 V input capacitor at each URB module (URB datasheet Cin).




**BOM accessories (not on the PCB)**

| | | |
|---|---|---|
| | FU201 | 1 A 5x20 glass fuse, in F201 |
| | FU202 | 3 A 5x20 glass fuse, in F202 |
| | FC201 | clear cover for F201 |
| | FC202 | clear cover for F202 |
| | P201 | 4–pole cable plug for J201 |

**Design notes**

Loads: +3V3 about 0.4 A of the module's 1.35 A (30 %). 24V_LOGIC input about 0.3 A at full module load: 1 A fuse.

24V_IO: 5 V module 7.3 W + 15 V module 7 W at full load = 0.75 A, plus relay coils 0.21 A and sensors through the PTCs 1 A: 3 A fuse.

Input inrush I2t into the input capacitors stays below 1.1 A2s; the fuses are rated 4.26 A2s (1 A) and 12.5 A2s (3 A).

URB input 9–36 V (40 V continuous, 50 V for 1 s). SMBJ30A: 30 V stand–off above the 28.8 V maximum input, clamps 48.4 V at 12.4 A.

CTRL left open = module on. NC pins are not connected inside the module.

Two 24 V entries, protection, three isolated DC–DC modules

**Yantra Innovation – LCEC–IO**

Sheet: /Power/

File: 02_power.kicad_sch

***Title: LCEC–IO – 2 Power***

| | | |
|---|---|---|
| Size: A3 | Date: 2026–11–06 | **Rev: A** |
| KiCad E.D.A. 10.0.8 | | Id: 2/3 |

# 3 MCU — STM32F405VGT6, shared 25 MHz clock, SWD, UART2, heartbeat LED

51 signals used (pin map in docs/plan.md), PH1 is a test pad, the 30 pins of ports D and E are free for future use (no-connect flags). Firmware: HSE bypass, PLLM 25 / PLLN 336 / PLLP 2 = 168 MHz, IWDG enabled in option bytes.

**U301 power: 100 nF beside each VDD pin and VBAT, 4.7 uF bulk, VDDA through a ferrite, 2.2 uF on each VCAP**

X301 — 25 MHz

OE tied high = oscillator running. 22 R series resistors to PH0 (HSE bypass) and to the LAN9252 (sheet 4).

U301B STM32F405VGT6

VDDA / VREF+

U301A STM32F405VGT6

SWD (Cortex 10-pin): 1 VTref, 2 SWDIO, 3/5/9 GND, 4 SWCLK, 6 SWO, 7/8 not used, 10 NRST

SWD 2x5 1.27 mm

USART2: 1 TX (PA2), 2 RX (PA3), 3 GND

USART2

Test points: SPI1 (LAN9252), SPI2 (DAC), relay OE, analog clamp, MPG A/B and the spare pin PH1

| TP301 | TP302 | TP303 | TP304 | TP305 | TP306 | TP307 | TP308 | TP309 | TP310 | TP311 | TP312 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| ESC_SCK | ESC_MISO | ESC_MOSI | ESC_CS | DAC_SCK | DAC_MOSI | DAC_CS | RELAY_OE | AO_CLAMP | MPG_A | MPG_B | PH1_TP |

STM32F405VGT6 (LQFP-100), shared clock, SWD, UART, heartbeat LED

**Yantra Innovation — LCEC-IO**

Sheet: /MCU/
File: 03_mcu.kicad_sch

***Title: LCEC-IO — 3 MCU***

| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 3/3 |